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Кафедра «Архитектура и цифровые технологии»

Тема 5. Язык PHP и синтаксис

Автор: Мусаджанова Д.А.

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План:



5.1. Команды языка PHP.

5.2. Дайте подробную информацию об инструментах для работы с объектами в языке PHP

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🌍 PHP da yozilgan mashhur saytlar va loyihalar

1. Facebook (dastlabki bosqichda)

- Facebookning birinchi versiyasi **PHPda yozilgan**.
- Keyinchalik u **HipHop Virtual Machine (HHVM)** orqali optimallashtirildi.

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2. Wikipedia

- Dunyoning eng katta onlayn ensiklopediyasi.
- MediaWiki platformasi PHP'da yozilgan.

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3. WordPress

- Dunyodagi eng mashhur CMS (Content Management System).
- 40% dan ortiq saytlar **WordPress** asosida ishlaydi.

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4. Drupal va Joomla

- Katta hajmdagi saytlar uchun mo'ljallangan ochiq manbali CMS'lar.
- Har ikkalasi ham **PHP tilida yozilgan**.

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5. Yahoo!

- Yahoo'ning asosiy xizmatlari PHP'da ishlab chiqilgan.

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6. Mailchimp

- Mashhur email marketing platformasi.
- Backend qismi PHP yordamida qurilgan.

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7. Tumblr

- Mashhur blogging platformasi.
- Foydalanuvchilar soni yuz millionlab.



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8. Slack (ba'zi qismlari)

- Umumiy arxitektura boshqa texnologiyalar bilan, ammo server tomonida PHP ham qo'llangan.

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9. Flickr

- Mashhur foto almashish platformasi.

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10. Boshqa mashhur PHP asosidagi loyihalar

- **Baidu** (Xitoyning Google'i) — PHP'dan foydalanadi.
- **Canva** — dizayn platformasi, backendida PHP ham bor.
- **Etsy** — onlayn savdo platformasi.
- **Magento** — e-commerce (onlayn do'kon yaratish) tizimi.
- **Laravel, Symfony, CodeIgniter** kabi PHP freymvorklar asosida qurilgan ko'plab biznes loyihalar.

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5.1. Команды языка PHP

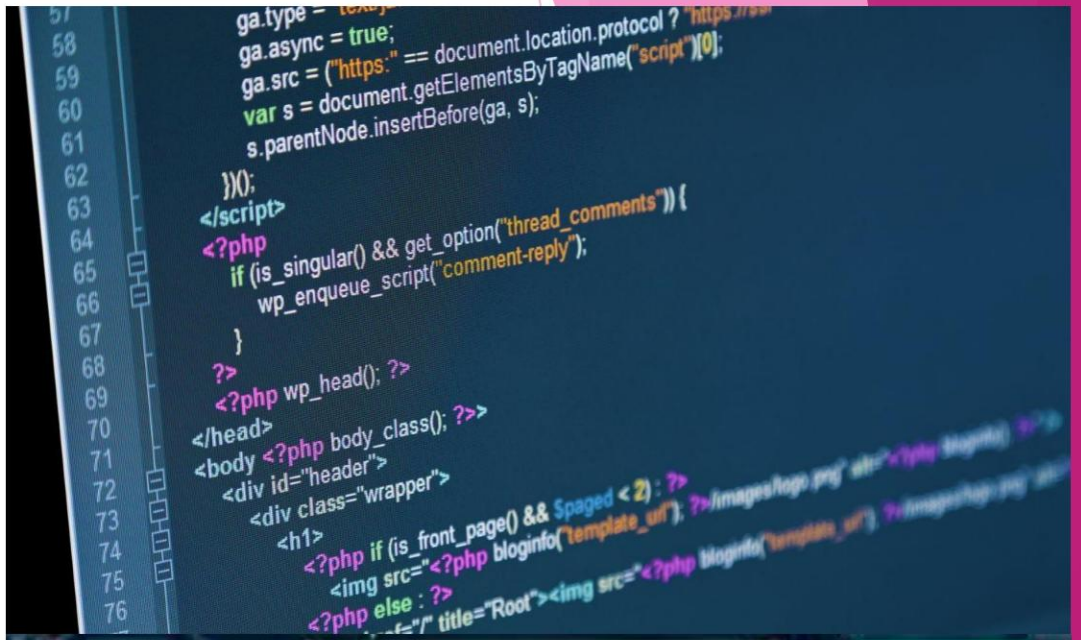
В PHP команды (или операторы) — это инструкции, выполняемые интерпретатором PHP. Каждая команда PHP обычно заканчивается точкой с запятой (;) .

Примеры основных команд PHP:

1. Отпустите команду

echo — выводит информацию (текст, переменную, число).

print похож на echo, за исключением того, что возвращает значение (0 или 1).





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пример простого кода php

php

Copy code

```php
<?php
echo "Salom, PHP!";
print "Bu ham chiqaradi.";
?>
```



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2. O‘zgaruvchilar bilan ishlash komandasi

PHPda $ belgisi bilan o‘zgaruvchi e’lon qilinadi.

php

Copy code

```php
<?php
$ism = "Ali";
$yosh = 21;
echo "Ism: $ism, Yosh: $yosh";
?>
```

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3. Arifmetik komandalar

- Qo‘shish: `+`
- Ayirish: `-`
- Ko‘paytirish: `*`
- Bo‘lish: `/`
- Qoldiq: `%`

php

Copy code

```php
<?php
$a = 10;
$b = 3;
echo $a + $b; // 13
echo $a % $b; // 1
?>
```

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php

```php
<?php
// example.php

// Sonlarni aniqlash
$a = 12;
$b = 5;

// Qo'shish
$sum = $a + $b;           // 17

// Ayirish
$diff = $a - $b;          // 7

// Ko'paytirish
$product = $a * $b;       // 60

// Bo'lish (float natija)
$quotient = $a / $b;      // 2.4

// Qoldiq (modulo)
$remainder = $a % $b;     // 2

// Kuchga ko'tarish (pow yoki **)
$power = $a ** 2;         // 144
$power2 = pow($b, 3);     // 125

// Inkrement va dekrement
$c = 10;
$c++;                     // endi $c = 11
$d = 10;
--$d;                     // endi $d = 9

// Kombinatsiyalangan ifoda
$result = ($a + $b) * ($c - $d) / $b; // misol: (12+5)*(11-9)/5 = 17*2/5 = 6.8

// Natijalarni chiqarish
echo "a = $a, b = $b\n";
echo "Qo'shish: a + b = $sum\n";
echo "Ayirish: a - b = $diff\n";
echo "Ko'paytirish: a * b = $product\n";
echo "Bo'lish: a / b = $quotient\n";
echo "Qoldiq: a % b = $remainder\n";
echo "Kuchga ko'tarish: a^2 = $power, b^3 = $power2\n";
echo "Inkrement: c = $c, Dekrement: d = $d\n";
echo "Kombinatsiyalangan ifoda natijasi: $result\n";

// Bo'lib bo'lganda 0 ga bo'linishni tekshirish (muhim)
$x = 8;
$y = 0;
if ($y == 0) {
    echo "Diqqat: 0 ga bo'lib bo'lmaydi — y = $y\n";
} else {
    echo "x / y = " . ($x / $y) . "\n";
}
?>
```

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php

```php
<?php
// example.php

// Sonlarni aniqlash
$a = 12;
$b = 5;

// Qo'shish
$sum = $a + $b;           // 17

// Ayirish
$diff = $a - $b;          // 7

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// Inkrement va dekrement
$c = 10;
$c++;                     // endi $c = 11
$d = 10;
--$d;                     // endi $d = 9

// Kombinatsiyalangan ifoda
$result = ($a + $b) * ($c - $d) / $b; // misol: (12+5)*(11-9)/5 = 17*2/5 = 6.8

// Natijalarni chiqarish
echo "a = $a, b = $b\n";
echo "Qo'shish: a + b = $sum\n";
echo "Ayirish: a - b = $diff\n";
echo "Ko'paytirish: a * b = $product\n";
echo "Bo'lish: a / b = $quotient\n";
echo "Qoldiq: a % b = $remainder\n";
echo "Kuchga ko'tarish: a^2 = $power, b^3 = $power2\n";
echo "Inkrement: c = $c, Dekrement: d = $d\n";
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$x = 8;
$y = 0;
if ($y == 0) {
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} else {
    echo "x / y = " . ($x / $y) . "\n";
}
?>
```

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Saved memory full

Kod tahlili:

1. O‘zgaruvchilarni aniqlash:

```php
$a = 12;
$b = 5;
```

- `$a` qiymati 12 ga,
- `$b` qiymati 5 ga teng.

---

2. Arifmetik amallar

- **Qo‘shish:**

```php
$sum = $a + $b;   // 12 + 5 = 17
```

- **Ayirish:**

```php
$diff = $a - $b;  // 12 - 5 = 7
```

- **Ko‘paytirish:**

```php
$product = $a * $b;   // 12 * 5 = 60
```

- **Bo‘lish (kasr natija):**

```php
$quotient = $a / $b;  // 12 / 5 = 2.4
```

- **Qoldiq (modulo):**

```php
$remainder = $a % $b; // 12 % 5 = 2
```

- **Darajaga ko‘tarish:**

```php
$power = $a ** 2;     // 12^2 = 144
$power2 = pow($b, 3); // 5^3 = 125
```

---

3. Inkrement va dekrement

```php
$c = 10;
$c++;   // c = 11 (1 ga oshdi)

$d = 10;
--$d;   // d = 9  (1 ga kamaydi)
```

---

4. Kombinatsiyalangan ifoda

```php
$result = ($a + $b) * ($c - $d) / $b;
```

Hisoblaymiz:

- `(12 + 5) = 17`
- `(11 - 9) = 2`
- `17 * 2 = 34`
- `34 / 5 = 6.8`

👉 `$result = 6.8`

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5. Natijalarni chiqarish

php

Copy code

```php
echo "a = $a, b = $b\n";
echo "Qo'shish: a + b = $sum\n";
...
```

Konsolda quyidagi natija chiqadi:

makefile

Copy code

```makefile
a = 12, b = 5
Qo'shish: a + b = 17
Ayirish: a - b = 7
Ko'paytirish: a * b = 60
Bo'lish: a / b = 2.4
Qoldiq: a % b = 2
Kuchga ko'tarish: a^2 = 144, b^3 = 125
Inkrement: c = 11, Dekrement: d = 9
Kombinatsiyalangan ifoda natijasi: 6.8
```

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6. Nolga bo‘lishni tekshirish

php

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```php
$x = 8;
$y = 0;

if ($y == 0) {
    echo "Diqqat: 0 ga bo'lib bo'lmaydi — y = $y\n";
} else {
    echo "x / y = " . ($x / $y) . "\n";
}
```

- `$y = 0` bo‘lgani uchun shart ishlaydi.
- Natija:

rust

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```
Diqqat: 0 ga bo'lib bo'lmaydi — y = 0
```

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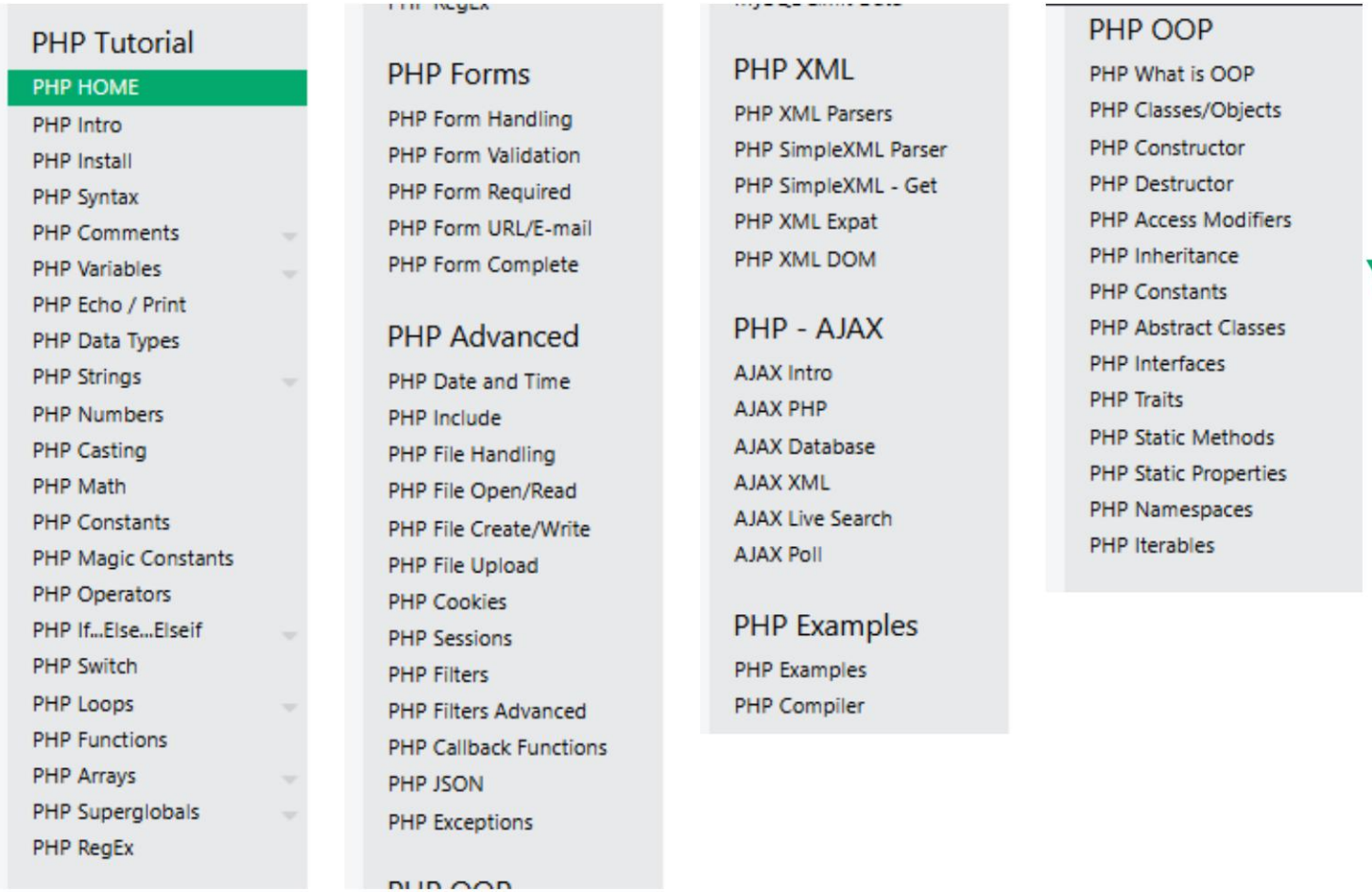
Umumiy xulosa:

Bu kod:

- Asosiy arifmetik amallarni (`+` , `-` , `*` , `/` , `%` , `**`) ko‘rsatib beradi.
- Inkrement (`++`) va dekrement (`--`) ishlashini amalda ko‘rsatadi.
- Murakkab ifoda qanday hisoblanishini misol bilan tushuntiradi.
- Eng muhimi — **0 ga bo‘lish xatosini oldini olish** uchun shart tekshiruvi (`if`) ishlatilgan.

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https://www.w3schools.com/php/default.asp





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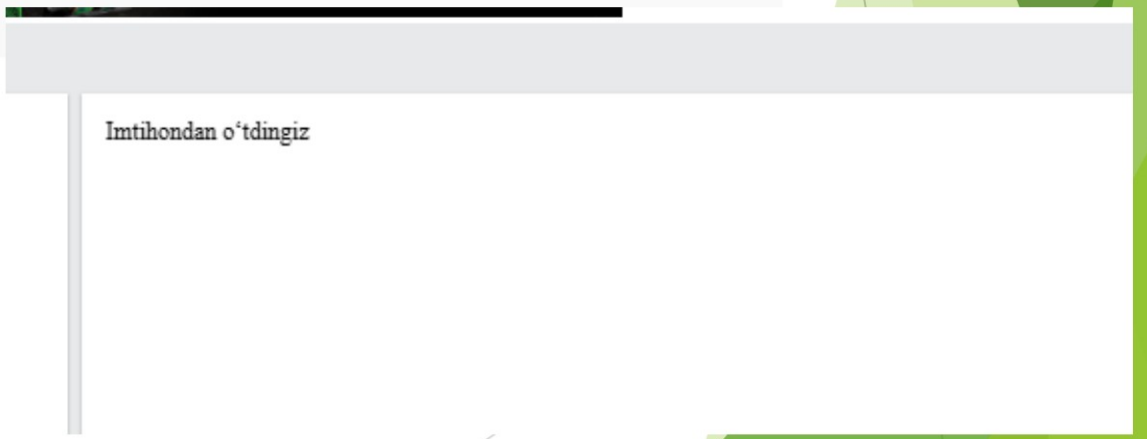
4. Shartli komandalar

php

Copy code

```php
<?php
$ball = 75;

if ($ball >= 60) {
    echo "Imtihondan o‘tdingiz";
} else {
    echo "Imtihondan o‘ta olmadingiz";
}
?>
```



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Tahlil:

1. `<?php ... ?>`

   Bu — PHP kodining boshlanish va tugash belgisi. PHP dasturlari shu belgilar orasida yoziladi.

2. `$ball = 75;`

   `$ball` — bu **o‘zgaruvchi**. Unga `75` qiymat berilgan.

   Ya’ni, talabaning imtihondan olgan bali **75** deb belgilangan.

3. `if ($ball >= 60) { ... }`

   - Bu **shart operatori**.
   - Tekshirilyapti: `$ball` 60 ga teng yoki undan katta bo‘lsa, ichidagi kod ishlaydi.
   - Bizning misolda `$ball = 75`, demak 75 ≥ 60 → shart **to‘g‘ri (true)**.

4. `echo "Imtihondan o‘tdingiz";`

   - Agar shart **to‘g‘ri** bo‘lsa, bu qator bajariladi.
   - Natija sifatida ekranga "**Imtihondan o‘tdingiz**" yozuvi chiqadi.

5. `else { echo "Imtihondan o‘ta olmadingiz"; }`

   - Agar shart **yolg‘on (false)** bo‘lsa, ya’ni `$ball < 60` bo‘lsa, ushbu qism bajariladi.
   - Ekranga "**Imtihondan o‘ta olmadingiz**" deb chiqadi.

```php
<?php
$ball = 75;

if ($ball >= 60) {
    echo "Imtihondan o‘tdingiz";
} else {
    echo "Imtihondan o‘ta olmadingiz";
}
?>
```

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5.2. PHP tilida ob’ekt vositalari (OOP – Object-Oriented Programming)

PHP faqat oddiy buyruqlar bilan emas, balki **ob’ektga yo‘naltirilgan dasturlash** (OOP) tamoyillari bilan ham ishlaydi. Bu yondashuv katta loyihalarda kodni tartibli, qayta ishlatiladigan va kengaytiriladigan qiladi.

OOP asosiy tushunchalari:

1. Class (sinflar)

- **Class** — obyekt yaratish uchun shablon.

php Copy code

```php
<?php
class Talaba {
    public $ism;
    public $yosh;

    function setData($ism, $yosh) {
        $this->ism = $ism;
        $this->yosh = $yosh;
    }

    function getInfo() {
        return "Ism: $this->ism, Yosh: $this->yosh";
    }
}
?>
```



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2. Object (obyektlar)

- Classdan obyekt yaratish uchun `new` kalit so'zi ishlatiladi.

php Copy code

```php
<?php
$talaba1 = new Talaba();
$talaba1->setData("Ali", 20);
echo $talaba1->getInfo();
?>
```



---

3. Properties (xususiyatlar) va Methods (metodlar)

- **Properties** — obyektning xossalari (`public $ism;`)
- **Methods** — obyektning funksiyalari (`function getInfo() { ... }`)

---

4. Access Modifiers (kirish darajalari)

- `public` — hamma joyda ishlatish mumkin
- `private` — faqat class ichida foydalaniladi
- `protected` — class ichida va undan meros olganlarda ishlaydi

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4. Access Modifiers (kirish darajalari)

- `public` — hamma joyda ishlatish mumkin
- `private` — faqat class ichida foydalaniladi
- `protected` — class ichida va undan meros olganlarda ishlaydi

php

Copy code

```php
class BankHisobi {
    private $balans = 0;

    public function pulQosh($miqdor) {
        $this->balans += $miqdor;
    }

    public function balansniKorish() {
        return $this->balans;
    }
}

$hisob = new BankHisobi();
$hisob->pulQosh(1000);
echo $hisob->balansniKorish(); // 1000
```



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5. Inheritance (Meros olish)

Bir classdan boshqasiga xususiyat va metodlarni meros qilib olish.

php

Copy code

```php
class Inson {
    public $ism;
    function __construct($ism) {
        $this->ism = $ism;
    }
}

class Oqituvchi extends Inson {
    public $fan;
    function __construct($ism, $fan) {
        parent::__construct($ism);
        $this->fan = $fan;
    }
    function getInfo() {
        return "$this->ism $this->fan fanidan dars beradi";
    }
}

$oqituvchi = new Oqituvchi("Dilshod", "Matematika");
echo $oqituvchi->getInfo();
```

